

OS Project Demonstration Guide

Step 1 — Process Scheduling

Commands:

```
sched add P1 --burst=6 --arrival=0 --priority=2
sched add P2 --burst=8 --arrival=1 --priority=1
sched add P3 --burst=7 --arrival=2 --priority=3
sched add P4 --burst=3 --arrival=3 --priority=2
```

```
sched list
```

Run algorithms:

```
sched run --algo=fdfs
sched run --algo=rr --quantum=2
sched run --algo=priority
sched compare
```

Explain to Sir:

"Here six scheduling algorithms can be compared together, including waiting time and CPU utilization."

Step 2 — Memory Management

```
mem init --frames=16
mem alloc P1 --pages=3 --algo=firstfit
mem alloc P2 --pages=4 --algo=bestfit
mem alloc P3 --pages=2 --algo=worstfit
```

```
mem show
mem free P2
mem show
```

Page Replacement:

```
mem pagefault --algo=fifo --frames=3 --ref=1,2,3,4,1,2,5,1,2,3,4,5
mem pagefault --algo=lru --frames=3 --ref=1,2,3,4,1,2,5,1,2,3,4,5
mem compare --frames=3 --ref=1,2,3,4,1,2,5,1,2,3,4,5
```

Explain:

"FIFO, LRU, Optimal and Clock algorithms can be compared using the same reference string."

Step 3 — Virtual File System

```
fs tree
fs mkdir /home/user/documents
fs touch /home/user/documents/report.txt
fs write /home/user/documents/report.txt "This is my OS project"
fs cat /home/user/documents/report.txt
fs chmod /home/user/documents/report.txt 755
fs stat /home/user/documents/report.txt
fs inode
fs fat /home/user/documents/report.txt
fs tree
```

Explain:

"Similar to a real ext filesystem, inode table, FAT cluster chain and permissions are implemented."

Step 4 — Disk Scheduling

```
disk init --tracks=200 --head=53
disk add 98 183 37 122 14 124 65 67
```

```
disk run --algo=fcfs
disk run --algo=scan
disk compare
```

Explain:

"Disk head movement visualization shows total seek distance comparison among algorithms."

Step 5 — Deadlock (Banker's Algorithm)

```
deadlock init --processes=P0,P1,P2 --resources=A,B,C
deadlock max P0 7 5 3
deadlock max P1 3 2 2
deadlock max P2 9 0 2
```

```
deadlock alloc P0 0 1 0
deadlock alloc P1 2 0 0
deadlock alloc P2 3 0 2
```

```
deadlock available 3 3 2
deadlock check
deadlock rag
```

Explain:

"Banker's Algorithm finds safe sequences step by step using Resource Allocation Graph."

Step 6 — Synchronization

```
sync dining --philosophers=5 --steps=15 --animate
sync producer --buffers=5 --producers=2 --consumers=2 --steps=15
sync reader --readers=3 --writers=2 --steps=12
sync semaphore --initial=1 --steps=10
```

Step 7 — Live Dashboard

```
dashboard
```

Explain:

"Real-time system monitor displays CPU, RAM, swap and running processes. Exit using Ctrl+C."

Common Viva Questions

1. What technologies were used?

Answer: Python, Rich library for TUI and prompt_toolkit for interactive shell.

2. How does scheduling work?

Answer: Explain using the generated Gantt chart.

3. What is Banker's Algorithm?

Answer: Show the safe sequence table.

4. Why is page replacement necessary?

Answer: Compare page fault counts.

5. How was the file system implemented?

Answer: Demonstrate inode table and FAT structure.